



Question Set

Question 1

Consider a three-asset risk portfolio with equity, rates, and credit-spread factor returns. Define the state variables (expected returns, covariance matrix, factor exposures, constraints, and turnover penalties) and derive the mean-variance objective and KKT system. State the measure, filtration, and tradability assumptions.

$$\min_w \frac{1}{2} w^\top \Sigma w - \lambda \mu^\top w \quad \text{s. t.} \quad \mathbf{1}^\top w = 1$$

Question 2

For a three-asset risk portfolio with equity, rates, and credit-spread factor returns, explain which probability measure is natural for pricing. Write the density-process logic needed if the starting model is under the physical measure.

Question 3

Identify the discounted tradable or density-adjusted process in a three-asset risk portfolio with equity, rates, and credit-spread factor returns. Prove the martingale condition and name one modelling choice that would break it.

Question 4

Using 240 daily returns for three assets with non-zero correlations, construct a two-step numerical calculation. Show intermediate state values before discounting or aggregating.

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Question 5

For a three-asset risk portfolio with equity, rates, and credit-spread factor returns, compute and interpret factor-neutral or beta-neutral portfolio weights. Explain the hedge sign and the unit of each sensitivity.

Question 6

Suppose the desk observes factor betas and covariance estimated from returns. Write the calibration objective, the weights you would use, and the parameter-identifiability checks.

Question 7

State the terminal and boundary conditions for a three-asset risk portfolio with equity, rates, and credit-spread factor returns. Explain how budget, leverage, no-short, and concentration constraints prevents economically impossible prices.

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Question 8

Design a Monte Carlo or lattice estimator for a three-asset risk portfolio with equity, rates, and credit-spread factor returns. Specify portfolio PnL distribution under multivariate normal shocks, the estimator, the discounting rule, and the standard error.

Question 9

A junior quant implements a three-asset risk portfolio with equity, rates, and credit-spread factor returns and gets a stable but wrong answer. Explain how optimizing on unstable sample means without regularization creates the error and design a diagnostic test.

Question 10

Write an NumPy algorithm for a three-asset risk portfolio with equity, rates, and credit-spread factor returns. Include inputs, outputs, diagnostics, and one benchmark case.

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Question 11

Starting from the payoff definition of a three-asset risk portfolio with equity, rates, and credit-spread factor returns, derive a replication, dynamic-programming, or conditional-expectation representation without skipping the economic argument.

Question 12

Create a stressed-market version of 240 daily returns for three assets with non-zero correlations. Recompute the value or risk metric and explain which term dominates the change.

Question 13

Use a second-order Taylor approximation for a three-asset risk portfolio with equity, rates, and credit-spread factor returns. Separate linear exposure, convexity exposure, cross terms if relevant, and residual model risk.

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Question 14

Show how a change of numeraire or change of measure would simplify one expectation in a three-asset risk portfolio with equity, rates, and credit-spread factor returns. Identify what becomes a martingale after the change.

Question 15

Give a proof-style answer showing when the wealth process for a three-asset risk portfolio with equity, rates, and credit-spread factor returns is self-financing. Explain why predictable trading strategies matter.

Question 16

Market quotes for factor betas and covariance estimated from returns are noisy. Propose a robust calibration workflow with initial guesses, constraints, residual plots, and out-of-sample checks.

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Question 17

For a three-asset risk portfolio with equity, rates, and credit-spread factor returns, compare the mathematical boundary condition with the trader's intuition. Give one limiting case where the answer should be obvious.

Question 18

Design a convergence study for portfolio PnL distribution under multivariate normal shocks. Specify path counts, time-step refinement, confidence intervals, and the exact quantity plotted.

Question 19

Find one way that optimizing on unstable sample means without regularization could pass a superficial unit test. Write a stronger test that would fail for the biased implementation.

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Question 20

Explain a three-asset risk portfolio with equity, rates, and credit-spread factor returns and code the core routine. Give the derivation outline, pseudocode, and three numerical sanity checks.